

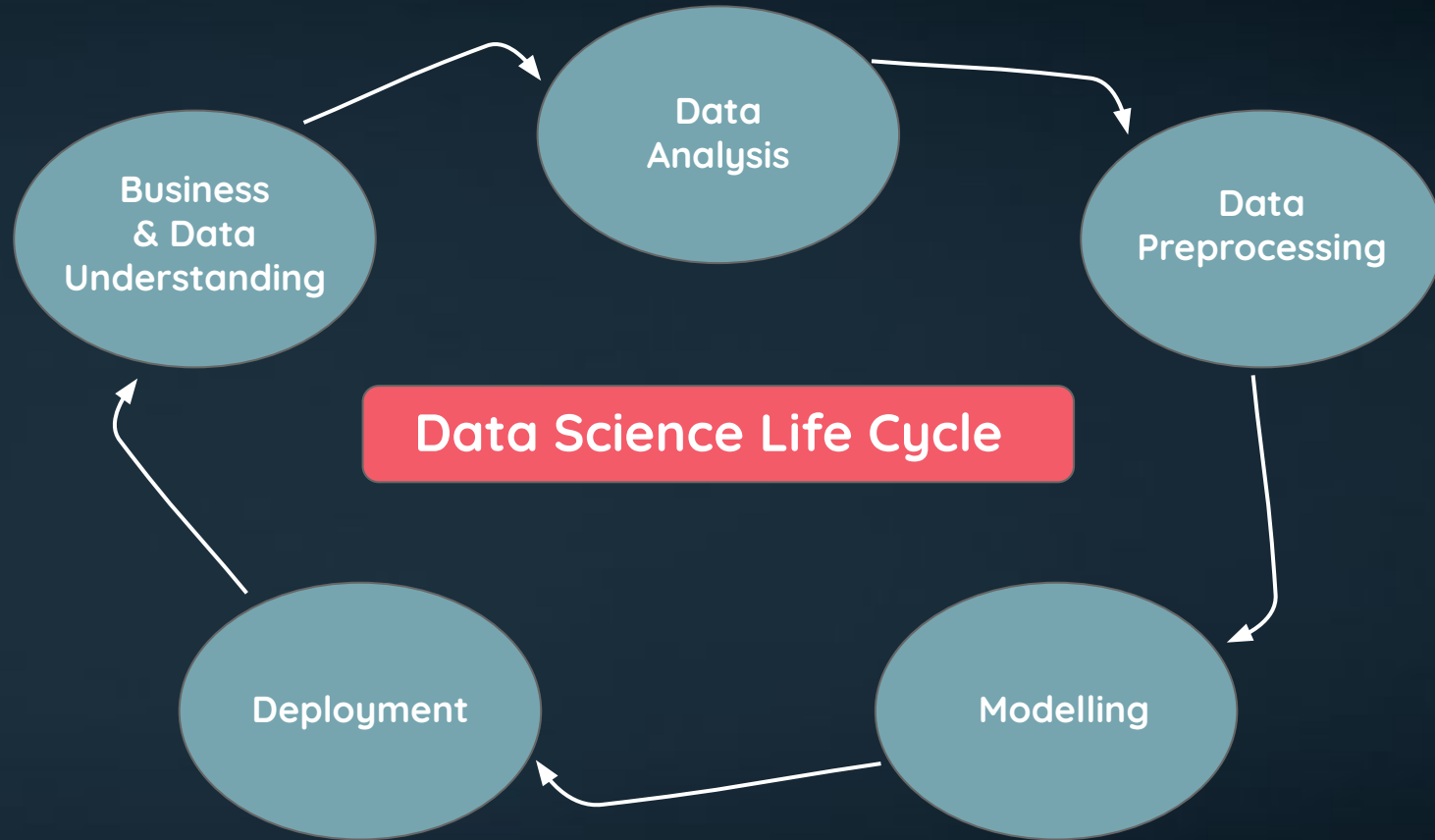
Intro to Data Analysis



Agenda

- Data Science Life Cycle.
- What is Data Analysis ?
- What is Data ?
- Data Types
- Data Representation
- Data Analysis Types
- Exploratory Data Analysis (EDA) Steps

Data Science Lifecycle



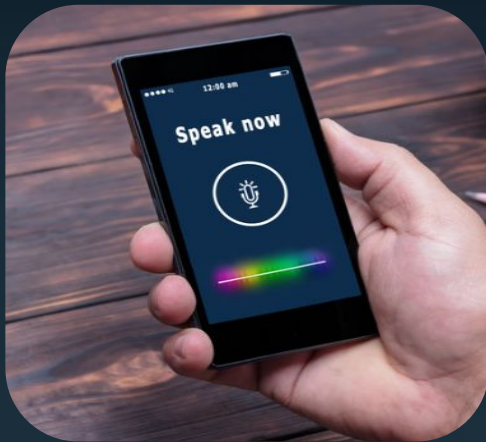
What is Data Analysis ?

What is Data Analysis ?

A process of examining data with the goal of discovering useful information, extract insights and informing conclusions to support decision-making

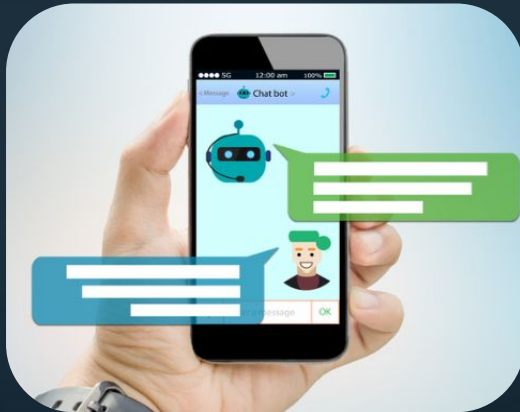


What is Data ?



	age (n)	job (c)	marital (c)	education (c)	balance (n)	housing (c)
0	30	unemployed	married	primary	1787	no
1	33	services	married	secondary	4789	yes
2	35	management	single	tertiary	1350	yes
3	30	management	married	tertiary	1476	yes
4	59	blue-collar	married	secondary	0	yes
5	35	management	single	tertiary	747	no

Data



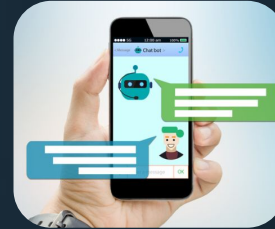
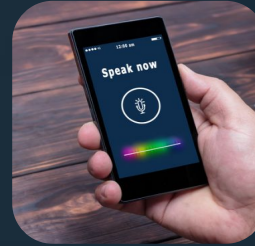
Data

Structured Data

	age (n)	job (c)	marital (c)	education (c)	balance (n)	housing (c)
0	30	unemployed	married	primary	1787	no
1	33	services	married	secondary	4789	yes
2	35	management	single	tertiary	1350	yes
3	30	management	married	tertiary	1476	yes
4	59	blue-collar	married	secondary	0	yes
5	35	management	single	tertiary	747	no



Unstructured Data



Employee	Position	Level	Salary	# supervision
Ahmad	HR Specialist	A	6500	2
Mohamed	Accountant	B	7500	1
Sayed	Lawyer	C	8450	0
Alaa	Engineer	B	9150	3
Adel	Trainer	C	4757	0
Osama	Sales	A	7546	2

**Structured
Data**

Nominal

Ordinal

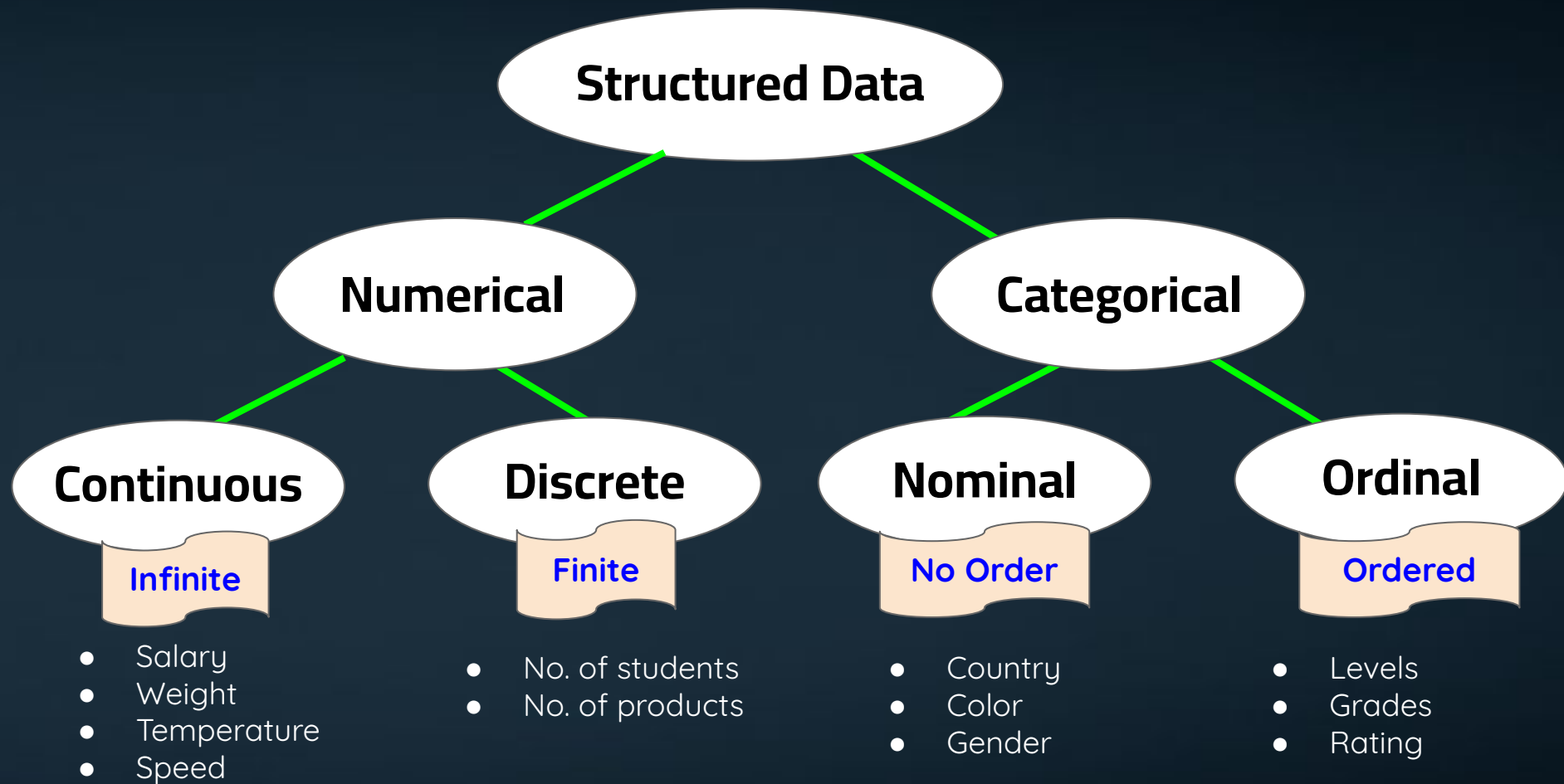
Continuous

Discrete

Categorical

Numerical

Data Types



Employee	Position	Level	Salary	# supervision
Ahmad	HR Specialist	A	6500	2
Mohamed	Accountant	B	7500	1
Sayed	Lawyer	C	8450	0
Alaa	Engineer	B	9150	3
Adel	Trainer	C	4757	0
Osama	Sales	A	7546	2

Analysis

- Relationship

Bivariate

Univariate

- Min
- Max
- Average

Statistics

Visualization

Data Representation

Employee	Position	Level	Salary	# supervision
Ahmad	HR Specialist	A	6500	2
Mohamed	Accountant	B	7500	1
Sayed	Lawyer	C	8450	0
Alaa	Engineer	B	9150	3
Adel	Trainer	C	4757	0
Osama	Sales	A	7546	2

Linear Algebra

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Representation

Mathematical

One
Collection

Matrix

In Python

Array

DataFrame

Python
Data Analysis

Numpy

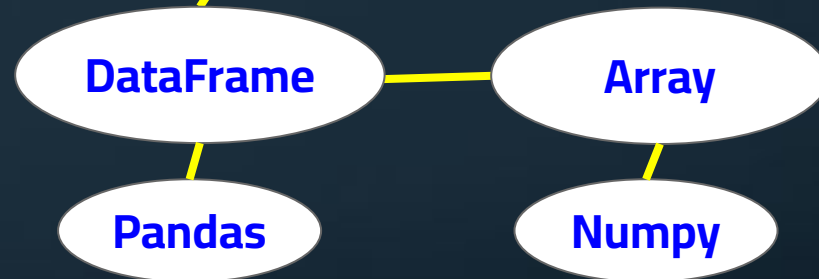
Pandas

Statistics

Position	Level	Salary	# supervision
HR Specialist	A	6500	2
Accountant	B	7500	1
Lawyer	C	8450	0
Engineer	B	9150	3
Trainer	C	4757	0
A	A	7546	2

Python Data Analysis

Employee	Position	Level	Salary	# supervision
Ahmad	HR Specialist	A	6500	2
Mohamed	Accountant	B	7500	1
Sayed	Lawyer	C	8450	0
Alaa	Engineer	B	9150	3
Adel	Trainer	C	4757	0
Osama	Sales	A	7546	2



Data Analysis Types

Types of Data Analysis

- Descriptive Analysis

> describe or summarize quantitative data by presenting statistics.

- Example:

*the distribution of sales across a group of employees and the **min/average/max/total** sales per employee.*

Descriptive Analysis answers the question, “What happened?”

Types of Data Analysis

- Diagnostic Analysis

> If the descriptive analysis determines the “what”, diagnostic analysis determines the “why”

- Example:

the sales increased and dropped several times during 10 years. Why ?

***Diagnostic Analysis
answers the question,
“Why did it happen?”***

Types of Data Analysis

- Predictive Analysis

> If we understood the past, we would predict the future

- Example:

Can we predict the sales for the next year ?

Predictive Analysis answers the question, “What might happen in the future?”

Types of Data Analysis

- Prescriptive Analysis

> If we predicted the future, How to be ready for it ?

- Example:

Can we make profit / avoid loss ?

Prescriptive Analysis answers the question, “What should we do?”

[More from this article \(Coursera\)](#)

Types of Data Analysis

Descriptive

“What happened?”

Diagnostic

“Why did it happen?”

Predictive

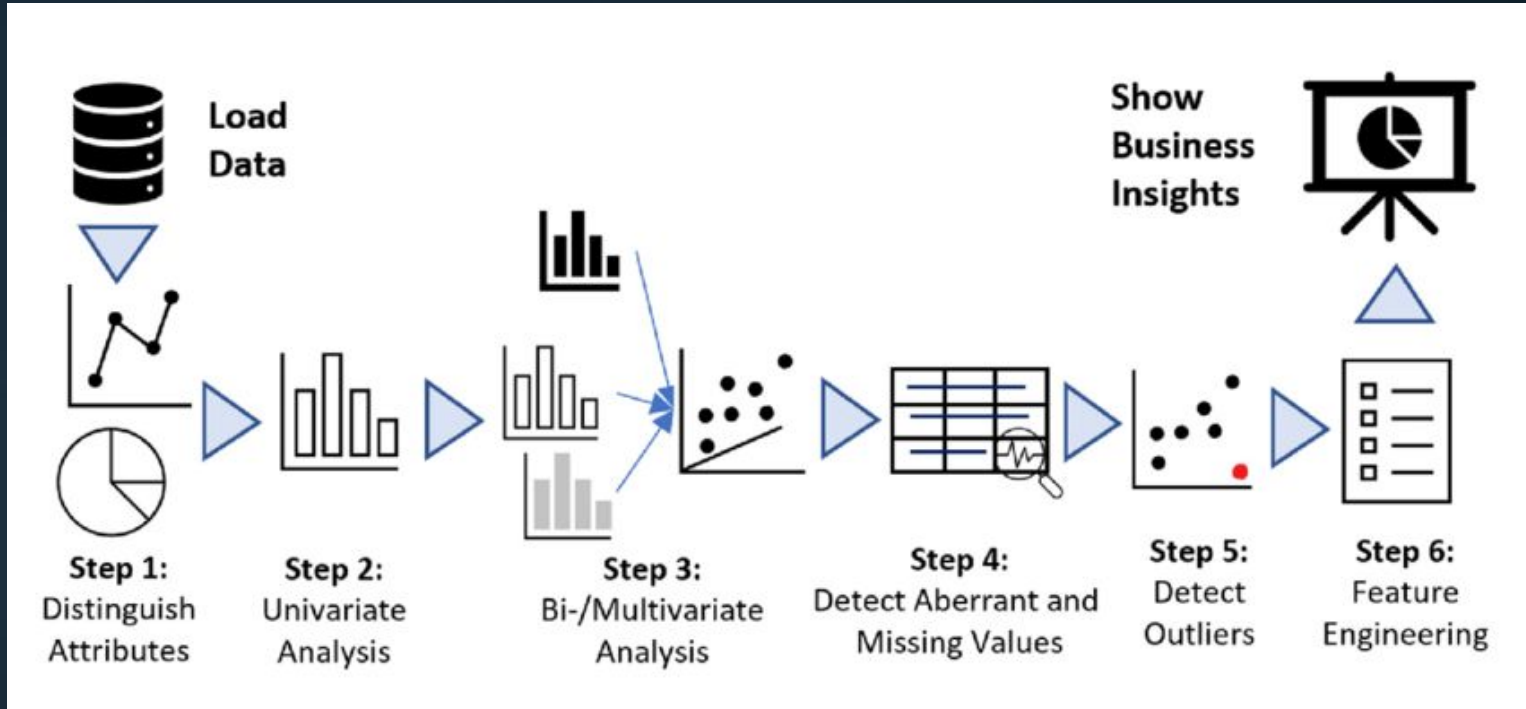
“What might happen in the future?”

Prescriptive

“What should we do?”

Exploratory Data Analysis

Exploratory Data Analysis (EDA) steps



[Source: Research Gate](#)



Done

